

Sourabh Sharma

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SUMMARY

Results-driven Software Engineer specializing in Python, FastAPI, and AI/ML integrations. Proven track record of architecting scalable backend systems, automating complex workflows, and migrating synchronous architectures to high-performance asynchronous environments. Expertise in developing microservices, deploying deep learning models (YOLO, SAM, TTS/STT), and implementing real-time event-driven communication (MQTT, RabbitMQ, EMQX).

EXPERIENCE

Predusk Technology Pvt Ltd

Associate Software Developer

Jaipur, India

Nov 2025 – Present

– Workplaceslm (RAG Chatbot & AI Services):

- Overhauled real-time chat architecture by migrating raw WebSockets to an MQTT broker (EMQX), developing a dedicated FastAPI microservice with JWT authentication for stable, persistent connections.
- Eliminated FastAPI event loop blocking by migrating synchronous `psycopg2` and `MinIO` SDK to `asyncpg` and `aioboto3`.
- Integrated Celery background tasks to handle compute-heavy deep-search operations, utilizing MQTT to stream search results directly to users in real-time, eliminating HTTP polling.
- Upgraded the deep-research AI pipeline from Agno to LangGraph, implementing a robust multi-agent architecture.
- Developed custom FastAPI microservices for voice integration: a Text-to-Speech (TTS) engine using Kokoro and a Speech-to-Text (STT) service using Faster-Whisper.

– DADP (Data Annotation and Delivery Platform):

- Replaced unstable Server-Sent Events (SSE) with a RabbitMQ and MQTT-based real-time event streaming microservice, ensuring reliable progress tracking for long-running Celery tasks.
- Architected and deployed a GPU-accelerated FastAPI microservice utilizing SAM 2.1 (Hiera Large) to dynamically generate polygon coordinates based on bounding box or click prompts for image annotation.
- Developed an automated video tracking microservice using YOLO11x (detection) and YOLO11x-cls (appearance embeddings) coupled with the BoT-SORT algorithm to auto-track objects across frames, minimizing manual annotator effort.

Predusk Technology Pvt Ltd

Data Analyst Trainee (AI & ML Engineering Team)

Jaipur, India

Jan 2025 – Oct 2025

– Oceanmind (Web Automation & Scraping Platform):

- Engineered a scalable automation engine for cruise quotations, replacing manual tasks. Migrated databases from SQLite/JSON to PostgreSQL and MinIO.
- Migrated web scraping architecture from Selenium to Playwright. Optimized browser orchestration by replacing per-request full browser instances with isolated browser contexts, drastically reducing RAM consumption and increasing speed.
- Integrated Google Gemini API to dynamically parse and structure unstructured special requests from customer comments.

PERSONAL PROJECTS

Multilingual TTS Microservice | FastAPI, Streamlit, Kokoro TTS, Python, Docker

- Designed and deployed a standalone Text-to-Speech microservice utilizing the Kokoro model, supporting dynamic language selection and multi-voice generation.
- Built an intuitive Streamlit frontend for seamless user interaction and containerized the application using Docker for consistent, environment-agnostic deployment.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

AI & Machine Learning: LLMs & RAG (LangGraph, Gemini API, vLLM), Vision (YOLO11x, SAM 2.1, BoT-SORT, OCR), Audio (Kokoro TTS, Faster-Whisper)

Backend Engineering: FastAPI, Celery, RESTful APIs, JWT

Frontend Development: Next.js, Streamlit, HTML/CSS

Databases & Storage: PostgreSQL (`asyncpg`, `psycopg2`), Redis, SQLite, MinIO, S3 Object Storage

Messaging & Real-Time: RabbitMQ, MQTT (EMQX), WebSockets

DevOps & Automation: Docker, Linux/VM Deployments, Git/GitHub, Playwright, Selenium

EDUCATION

SSG Pareek College (University of Rajasthan)

Bachelor of Computer Applications (BCA) | 66.6%

Jaipur, India

2022 – 2025